prometheus / prometheus Public

Code Issues 505 Pull requests 256 Discussions Actions

358 Branches 529 Tags Go to file Go to file Code

roidelapluie Merge pull request #18592 from roidelapluie/roidelapluie/rw-...

16034ec · 3 days ago

.github	Enable govulncheck in CI (#18...	last week
cmd	Merge pull request #18575 fro...	3 days ago
compliance	ci: add compliance linting step...	3 weeks ago
config	Merge pull request #18518 fro...	2 weeks ago
discovery	[ENHANCEMENT] AWS SD: A...	4 days ago
docs	Merge pull request #18575 fro...	3 days ago
documentation	fix(discovery): delete expired r...	28 days ago
internal/tools	chore: Go dependencies upda...	last month
model	rulefmt: warn when multi-doc...	3 weeks ago
notifier	fix(discovery): delete expired r...	28 days ago
plugins	feat(discovery): add Outscale ...	last month
prompb	Rename created timestamp to...	last month
promql	PromQL: Promote duration ex...	last week
rules	rulefmt: warn when multi-doc...	3 weeks ago
schema	Remove copyright date from h...	3 months ago
scrape	discovery, scrape: Use backof...	27 days ago
scripts	Enable govulncheck in CI (#18...	last week
storage	Merge pull request #18586 fro...	3 days ago
template	Remove copyright date from h...	3 months ago

tracing	tracing: add regression test fo...	3 weeks ago
tsdb	tsdb: Reduce test flakiness (#...	3 days ago
util	Merge pull request #18575 fro...	3 days ago
web	PromQL: Promote duration ex...	last week
.dockerignore	Fix dockerignore sync (#18488)	3 weeks ago
.gitattributes	Add OpenAPI 3.2 specification...	3 months ago
.gitignore	Add Go workspace for multi-m...	4 months ago
.golangci.yml	chore: fix emptyStringTest iss...	last month
.promu.yml	Support the aix/ppc64 compil...	last month
.yamllint	Add OpenAPI 3.2 specification...	3 months ago
AGENTS.md	Address review feedback	last month
CHANGELOG.md	changelog: add security entry ...	3 days ago
CLAUDE.md	Tell Claude to read it too	last month
CODEOWNERS	Add kgeckhart as agent maint...	2 weeks ago
CODE_OF_CONDUCT.md	Update link for referenced CN...	4 years ago
CONTRIBUTING.md	docs: Fix inaccuracies in root ...	4 months ago
Dockerfile	Add distroless Docker image v...	3 months ago
Dockerfile.distroless	Fix ARG scoping in Dockerfile....	3 months ago
LICENSE	Clean up license issues.	12 years ago
MAINTAINERS.md	Add kgeckhart as agent maint...	2 weeks ago
Makefile	Update common container arc...	3 weeks ago
Makefile.common	Enable govulncheck in CI (#18...	last week
NOTICE	UI: Speed up alerts/rules/... pa...	5 months ago
README.md	Fix typo in README	last week
RELEASE.md	adjust and simplify	last month
SECURITY-INSIGHTS.yml	chore: provide OSSF security i...	2 years ago

SECURITY.md	Fix markdown format (#18505)	3 weeks ago
VERSION	Release 3.11.2	2 weeks ago
go.mod	Merge pull request #18433 fro...	3 weeks ago
go.sum	Merge pull request #18433 fro...	3 weeks ago
go.work	chore: Go dependencies upda...	last month
renovate.json	chore(config): migrate config r...	2 months ago
ui-commits	promql: rename holt_winters t...	2 years ago

README Code of conduct Contributing Apache-2.0 license Security

Prometheus

Visit prometheus.io for the full documentation, examples and guides.

CI failing | container none | docker pulls 2G | go report A+ | openssf best practices passing | govulncheck passing | openssf scorecard 8 | CLOMonitor Report A | oss-fuzz fuzzing

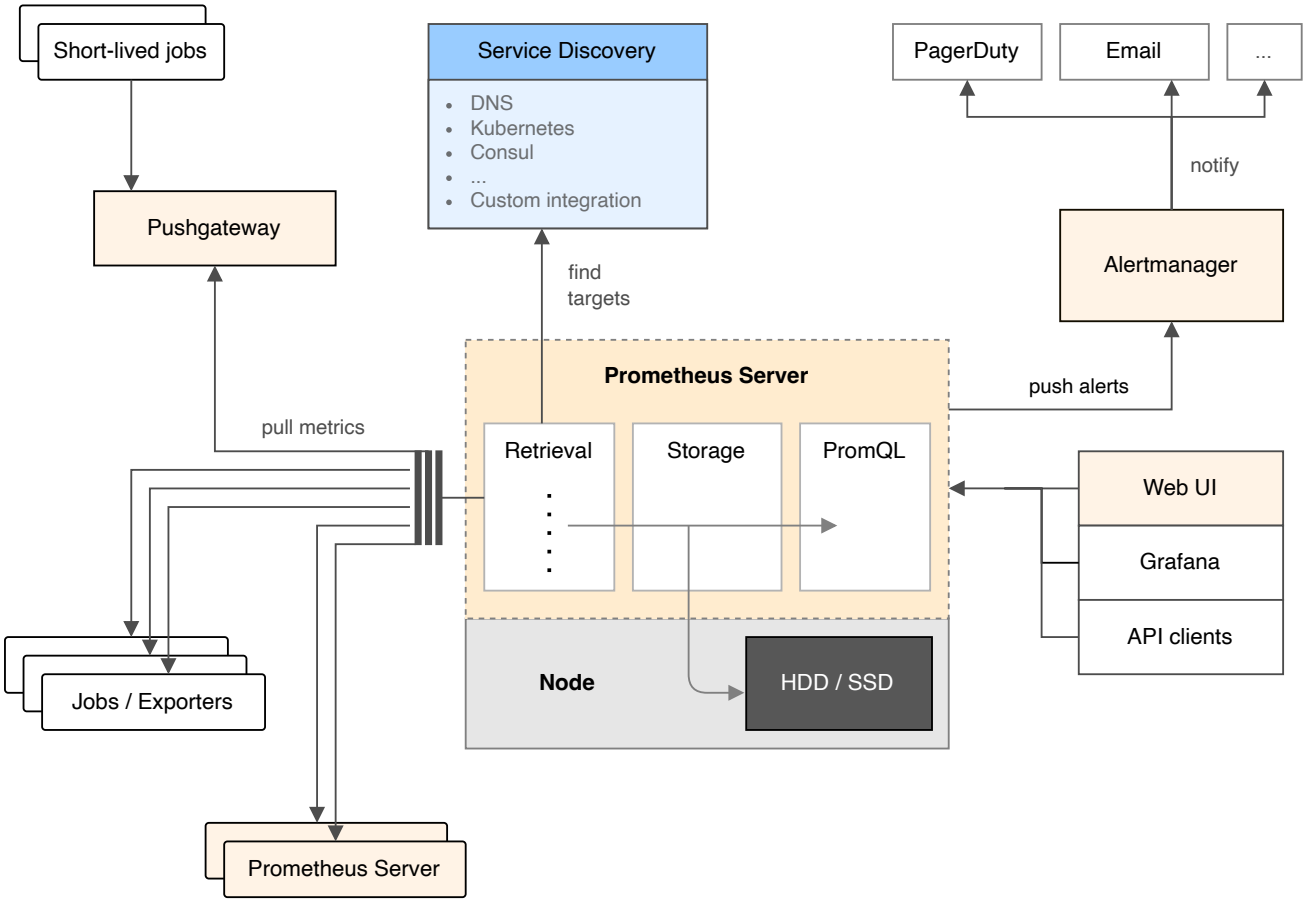
Prometheus, a Cloud Native Computing Foundation project, is a systems and service monitoring system. It collects metrics from configured targets at given intervals, evaluates rule expressions, displays the results, and can trigger alerts when specified conditions are observed.

The features that distinguish Prometheus from other metrics and monitoring systems are:

- A **multi-dimensional** data model (time series defined by metric name and set of key/value dimensions)
- PromQL, a **powerful and flexible query language** to leverage this dimensionality
- No dependency on distributed storage; **single server nodes are autonomous**
- An HTTP **pull model** for time series collection
- **Pushing time series** is supported via an intermediary gateway for batch jobs
- Targets are discovered via **service discovery** or **static configuration**
- Multiple modes of **graphing and dashboarding support**

- Support for hierarchical and horizontal **federation**

Architecture overview



Install

There are various ways to install Prometheus.

Precompiled binaries

Precompiled binaries for released versions are available in the *download* section on prometheus.io. Using the latest production release binary is the recommended way to install Prometheus. See the Installing chapter in the documentation for all the details.

Docker images

Docker images are available on Quay.io or Docker Hub.

You can launch a Prometheus container for trying it out with

```
docker run --name prometheus -d -p 127.0.0.1:9090:9090 prom/prometheus
```

Prometheus will now be reachable at http://localhost:9090/.

Building from source

To build Prometheus from source code, you need:

- Go: Version specified in go.mod or greater.
- NodeJS: Version specified in .nvmrc or greater.
- npm: Version 10 or greater (check with `npm --version` and here).

Start by cloning the repository:

```
git clone https://github.com/prometheus/prometheus.git
cd prometheus
```

You can use the `go` tool to build and install the `prometheus` and `promtool` binaries into your `GOPATH`:

```
go install github.com/prometheus/prometheus/cmd/...
prometheus --config.file=your_config.yml
```

However, when using `go install` to build Prometheus, Prometheus will expect to be able to read its web assets from local filesystem directories under `web/ui/static`. In order for these assets to be found, you will have to run Prometheus from the root of the cloned repository. Note also that this directory does not include the React UI unless it has been built explicitly using `make assets` or `make build`.

An example of the above configuration file can be found here.

You can also build using `make build`, which will compile in the web assets so that Prometheus can be run from anywhere:

```
make build
./prometheus --config.file=your_config.yml
```

The Makefile provides several targets:

- *build*: build the `prometheus` and `promtool` binaries (includes building and compiling in web assets)
- *test*: run the tests
- *test-short*: run the short tests
- *format*: format the source code
- *vet*: check the source code for common errors

- *assets*: build the React UI

Service discovery plugins

Prometheus is bundled with many service discovery plugins. You can customize which service discoveries are included in your build using Go build tags.

To exclude service discoveries when building with `make build`, add the desired tags to the `.promu.yml` file under `build.tags.all`:

```
build:
    tags:
        all:
            - netgo
            - builtinassets
            - remove_all_sd           # Exclude all optional SDs
            - enable_kubernetes_sd    # Re-enable only kubernetes
```

Then run `make build` as usual. Alternatively, when using `go build` directly:

```
go build -tags "remove_all_sd,enable_kubernetes_sd" ./cmd/prometheus
```

Available build tags:

- `remove_all_sd` - Exclude all optional service discoveries (keeps file_sd, static_sd, and http_sd)
- `enable_<name>_sd` - Re-enable a specific SD when using `remove_all_sd`

If you add out-of-tree plugins, which we do not endorse at the moment, additional steps might be needed to adjust the `go.mod` and `go.sum` files. As always, be extra careful when loading third party code.

Building the Docker image

You can build a docker image locally with the following commands:

```
make promu
promu crossbuild -p linux/amd64
make npm_licenses
make common-docker-amd64
```

The `make docker` target is intended only for use in our CI system and will not produce a fully working image when run locally.

Using Prometheus as a Go Library

Within the Prometheus project, repositories such as prometheus/common and prometheus/client-golang are designed as re-usable libraries.

The prometheus/prometheus repository builds a stand-alone program and is not designed for use as a library. We are aware that people do use parts as such, and we do not put any deliberate inconvenience in the way, but we want you to be aware that no care has been taken to make it work well as a library. For instance, you may encounter errors that only surface when used as a library.

Remote Write

We are publishing our Remote Write protobuf independently at buf.build.

You can use that as a library:

```
go get buf.build/gen/go/prometheus/prometheus/protocolbuffers/go@latest
```

This is experimental.

Prometheus code base

In order to comply with go mod rules, Prometheus release number do not exactly match Go module releases.

For the Prometheus v3.y.z releases, we are publishing equivalent v0.3y.z tags. The y in v0.3y.z is always padded to two digits, with a leading zero if needed.

Therefore, a user that would want to use Prometheus v3.0.0 as a library could do:

```
go get github.com/prometheus/prometheus@v0.300.0
```

For the Prometheus v2.y.z releases, we published the equivalent v0.y.z tags.

Therefore, a user that would want to use Prometheus v2.35.0 as a library could do:

```
go get github.com/prometheus/prometheus@v0.35.0
```

This solution makes it clear that we might break our internal Go APIs between minor user-facing releases, as breaking changes are allowed in major version zero.

React UI Development

For more information on building, running, and developing on the React-based UI, see the React app's README.md.

More information

- Godoc documentation is available via pkg.go.dev. Due to peculiarities of Go Modules, v3.y.z will be displayed as v0.3y.z (the y in v0.3y.z is always padded to two digits, with a leading zero if needed), while v2.y.z will be displayed as v0.y.z.
- See the [Community page](https://prometheus.io/community/) for how to reach the Prometheus developers and users on various communication channels.

Contributing

Releases 366

3.11.3 / 2026-04-27 Latest
3 days ago

+ 365 releases

Packages

No packages published

Contributors 1,137

+ 1,123 contributors

Languages

Go 87.4% TypeScript 11.3% Yacc 0.5% Shell 0.2% SCSS 0.2% CSS 0.1% Other 0.3%